

Notes for ‘Functions as rules, with domain, range and graph’

Important Ideas and Useful Facts:

- (i) **Functions:** A *function* f is a process or rule that takes an *input*, typically a real number x , and produces an *output*, also typically a real number $f(x)$. It is common to refer to the equation

$$y = f(x)$$

as representing f , and call x the *independent* variable and y the *dependent* variable.

- (ii) **Domain and range:** The *domain* of a function f is the set of inputs x such that the rule $f(x)$ is sensibly defined. The domain may also be subject to certain restrictions, particularly in the context of forming *inverse functions* (to be explained later) or in certain physical examples or applications.

The *range* of f is the set of outputs $f(x)$ that are produced as x varies over the domain.

For example, the interval $[0, \infty)$ is both the domain and range of the function f given by the rule $f(x) = \sqrt{x}$, as the rule cannot be applied to negative numbers, and the rule produces all nonnegative real numbers.

By contrast, consider the function g with rule $g(x) = 2 - \sqrt{x+3}$. The domain of g is $[-3, \infty)$, as the rule only applies to real numbers greater than or equal to -3 . The range of g is $(-\infty, 2]$, as the square root produces all nonnegative real numbers, which, when taken away from 2, produces all real numbers less than or equal to 2.

- (iii) **Graph of a Function:** When x and y are real numbers, the *graph* of f is the collection of ordered pairs (x, y) in the xy -plane such that $y = f(x)$. Most graphs of functions that you meet will be simple curves in the plane.

- (iv) **Linear functions:** A *linear function* has the form $y = mx + k$ where m and k are constants. Its graph is a line in the plane, where m is the slope and k is the y -intercept. The domain is the whole real line $\mathbb{R}$.

The range is again the whole real line $\mathbb{R}$, provided the slope m is nonzero.

If the slope m is zero, then the rule becomes $y = k$, representing a *constant function* taking only the value k , so that the range becomes $\{k\}$.

- (v) **The identity function:** The function $y = x$, that outputs the input without any change at all, is called the *identity function*. It is a special case of a linear function where the slope is 1 and the y -intercept is 0. Its graph is a line often referred to as the *diagonal* of the xy -plane, and has an important geometric role in forming *inverse functions* by reflection (see later).

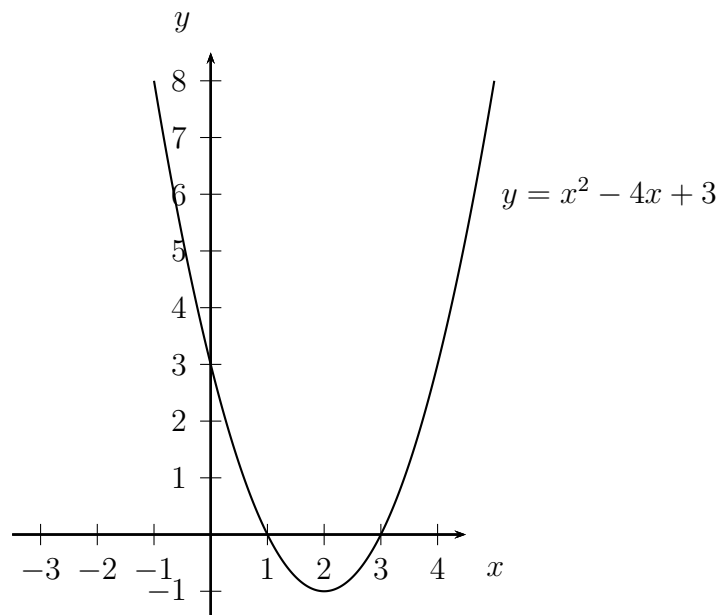
- (vi) **Quadratic functions:** A *quadratic function* has the form $y = ax^2 + bx + c$ where a , b and c are constants such that a is nonzero, and its graph is a parabola. The domain is the whole real line $\mathbb{R}$.

The range is an infinite interval, depending on whether the parabola faces upwards ($a > 0$) or downwards ($a < 0$), and on the position of the apex. The exact position of the apex and the range can be determined by rearranging the rule after completing the square.

For example, the rule $y = x^2 - 4x + 3$ corresponds to a parabola facing upwards, and can be rewritten, after completing the square, as

$$y = x^2 - 4x + 3 = x^2 - 4x + 4 - 4 + 3 = (x - 2)^2 - 1 ,$$

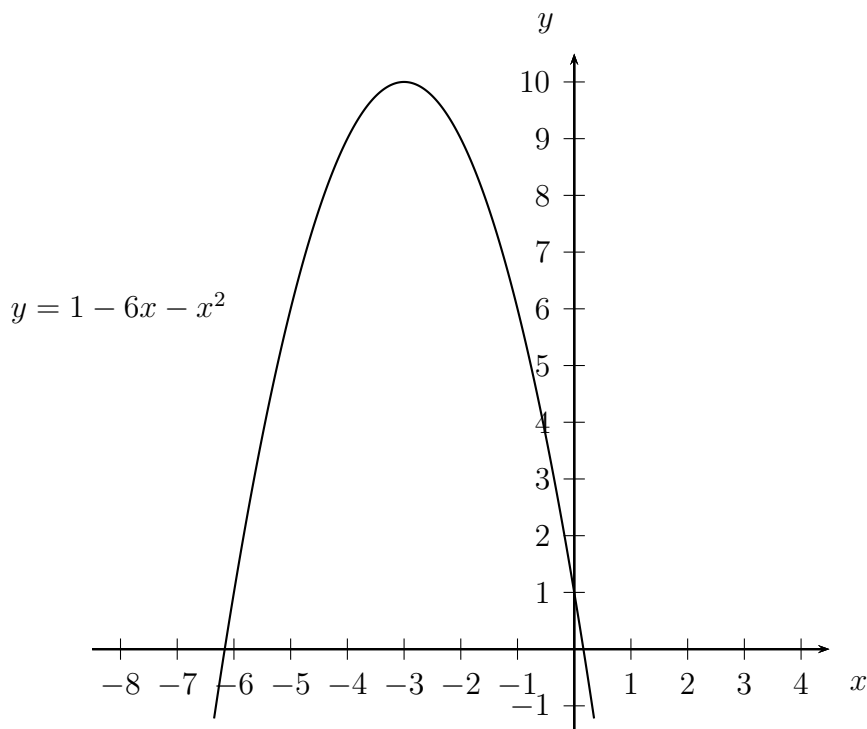
so that its apex is the point $(2, -1)$, and its range is the interval $[-1, \infty)$, since $(x - 2)^2 \geq 0$ for all x .



By contrast, the rule $y = 1 - 6x - x^2$ corresponds to a parabola facing downwards, and can be rewritten, again after completing the square, as

$$y = 1 - 6x - x^2 = 1 + 9 - 9 - 6x - x^2 = 10 - (x^2 + 6x + 9) = 10 - (x + 3)^2 .$$

The point $(-3, 10)$ is the apex, and the range is $(-\infty, 10]$, since $-(x + 3)^2 \leq 0$ for all x .



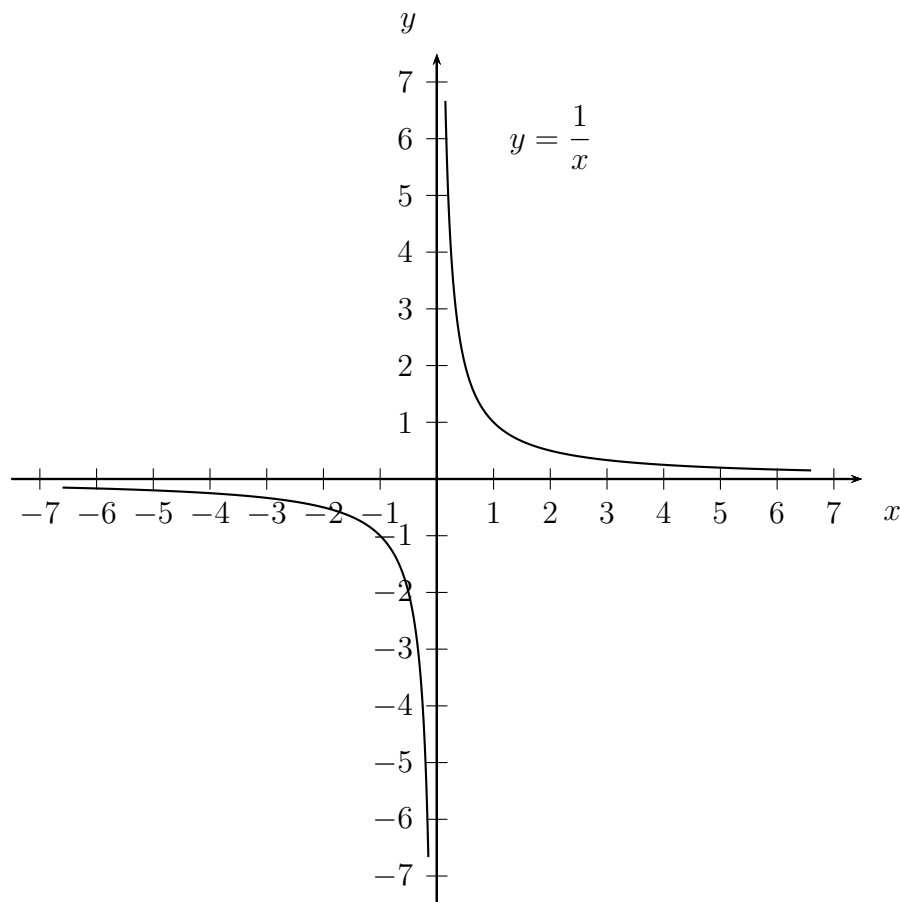
(vii) **Hyperbolas:** The simplest *hyperbola* is the graph of the function f with rule

$$y = f(x) = \frac{1}{x}.$$

For example,

$$f(1) = 1, \quad f(2) = \frac{1}{2}, \quad f\left(\frac{1}{2}\right) = 2, \quad f(-1) = -1, \quad f(-2) = -\frac{1}{2}, \quad f\left(-\frac{1}{2}\right) = -2.$$

The domain and range of f are both equal to $\mathbb{R} \setminus \{0\}$, the real line with the number 0 removed. The graph has two disconnected pieces, called *branches*, one in the first quadrant involving only positive numbers and their reciprocals, and the other in the third quadrant involving only negative numbers and their reciprocals.



The x -axis is a *horizontal asymptote* of the graph of f , approached by the curve both to the right and left as one moves further away from the origin. The y -axis is a *vertical asymptote*, approached by the curve as the inputs x get closer and closer to, but do not reach, zero.

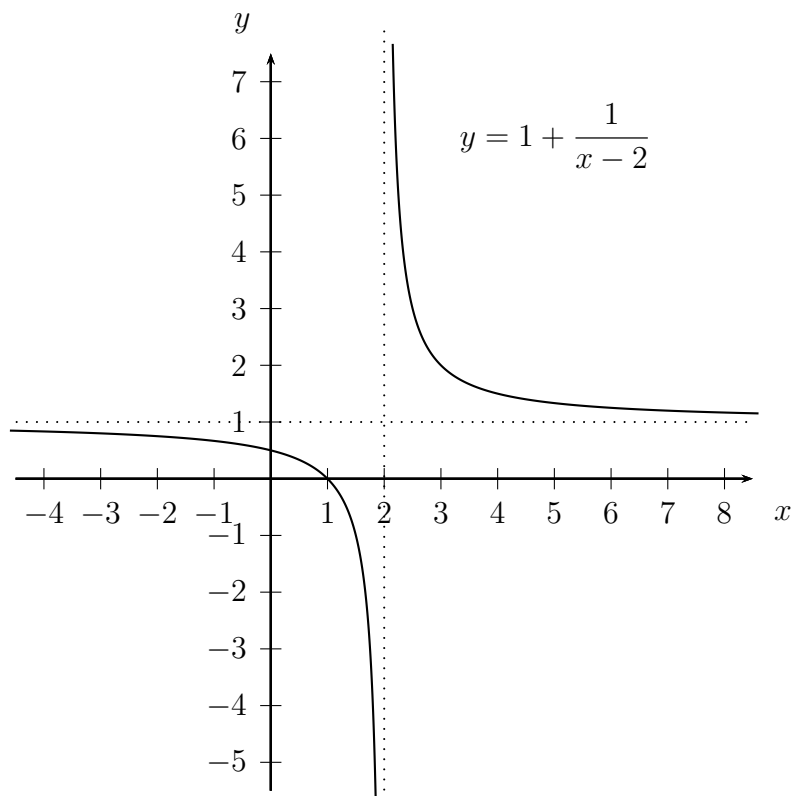
Consider, as a variation, the function g with rule

$$g(x) = 1 + \frac{1}{x-2} = \frac{x-1}{x-2}.$$

For example,

$$g(0) = \frac{1}{2}, \quad g(1) = 0, \quad g(3) = 2, \quad g(-1) = \frac{2}{3}, \quad g(-2) = \frac{3}{4}, \quad g\left(\frac{1}{2}\right) = \frac{1}{3}, \quad g\left(-\frac{1}{2}\right) = \frac{3}{5}.$$

However, $g(2)$ is undefined, and the domain of g is $\mathbb{R} \setminus \{2\}$, the real line with 2 removed. The fraction $\frac{1}{x-2}$ produces all nonzero real numbers, so, when this fraction is added to 1, the rule for g produces all real numbers other than 1, so the range is $\mathbb{R} \setminus \{1\}$.



The graph of g is another hyperbola, and obtained by shifting the graph of f horizontally by 2 units (the effect of having $x - 2$ in place of x in the denominator) and vertically by 1 unit (the effect of adding 1 to the fraction). These shifting effects move the horizontal asymptote to the line $y = 1$ and the vertical asymptote to the line $x = 2$.